Nikhil Kuppa

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PROFESSIONAL SUMMARY

Machine Learning and AI Engineer with an MS from Columbia University and 3.5+ years of experience building AI-driven solutions for startups, academia and industry alike. Skilled in deep learning, signal processing, stats, and LLM/GenAI pipelines, with expertise in Python, PyTorch, TensorFlow, and MATLAB. Experienced in translating research, ranging from physiological signals (ECG, EEG, rPPG) to video and text, into scalable, production-ready applications.

EDUCATION

Columbia University | MS in Biomedical Engineering [Dec '23] - **GPA: 3.88/4** **New York, NY**

Manipal University | B. Tech in Electronics and Communication Engineering [Jul '22] - **GPA: 8.38/10** **Manipal, India**

TECHNICAL SKILLS

Languages/Tools: Python, MATLAB, R, SQL, LangGraph, LangChain, RAG, AWS Bedrock, GCP, Claude Code

Specializations: Agentic AI, ML, Signal Processing, Image Processing, Statistical Analysis, Computer Vision, NLP, LLMs

Modalities: Image, Video, Audio, Text, Time Series, Clinical Trials (ADAM)

PROFESSIONAL EXPERIENCE

Novartis Pharmaceuticals – Contractor via Abacus Service Corporation **New York, NY**

Associate Data Scientist - Agentic AI **Nov 2025 – Present**

- Leading the development of an autonomous backend utilizing Claude Code to automate figure script and table generation for Clinical Study Reports (CSRs); engineered custom agentic skill-creators and evaluation frameworks that safeguard strict Statistical Analysis Plan compliance for global drug development submissions.
- Built an agentic RAG-based code generation system using LangGraph to fast-track Clinical Study Reporting - reducing time taken via manual effort by 99.17% while achieving an 85% success rate.

Parra Neural Engineering Lab, The City College of New York **New York, NY**

Research Engineer - Applied AI (Multimodal Data) **Aug 2024 – Jul 2025**

- Architected a Generative AI system using LLMs on 140,000+ educational videos to classify and analyze presence of learning principles via video files, transcripts and text embeddings. Scaled the pipeline with parallelization, used interactive visualizations to validate on the go.
- Engineered BIDS-compliant biomedical datasets (ECG, EEG, EOG, respiratory, eye-tracking), deployed them on an interactive Flask web app for data access – Website Link. Conducted statistical analyses to identify trends and anomalies. Published in Nature Scientific Data.
- Built real-time EEG analysis system with movie playback integration and processed large-scale video datasets with facial/pose landmark overlays using memory-efficient software architectures.

Heffner Biomedical Imaging Lab, Columbia University **New York, NY**

Research Engineer – Medical Image Analysis **Jun 2023 - Feb 2024**

- Executed Hidden Markov Measure-Field algorithm on the 3 Lung CT datasets to quantify emphysema and perform longitudinal analysis.
- Validated results via Bland-Altman analysis, computed CT Hounsfield Unit metrics and documented QC code.

Fourth Frontier Technologies Pvt. Ltd **Bangalore, India**

Intern - Digital Health Data Engineering & ML **Jan 2022 - Jul 2022**

- Overhauled workout activity insights by introducing a dynamic ranking system, and facilitated code deployment to production via AWS.
- Led the development and shipped a novel set of cardiac-based workout training insights (Heart Rate Zones) in collaboration with an MD.
- Engineered an extensive dataset of ECG and cardiac features for 2500+ users, with the goal of predicting cardiac age.

KEY PROJECTS

- **Non-Contact Camera-Based Vitals Tracking System** Built a production-grade rPPG API estimating HR, HRV, RR, and

BP from raw webcam video using dual algorithms (CHROM & POS), Welch's PSD, MediaPipe Face Mesh, and 32-feature pulse wave morphology for BP estimation; benchmarked 15 ML models for sclera-based jaundice detection (**best: R-G, GaussianNB, AUC 0.73, recall 0.79**) on a curated 1,200-sample dataset, and authored a proprietary data collection plan to train production-grade models; reduced API memory ~10x.

- **Phonocardiogram (PCG) Heart Sound Classifier** Developed a multi-stage ML classification pipeline using Wavelet Transform, achieving 0.97 AUC. Segmented S1/S2 heart sounds with wavelet-based peak detection, and deployed end-to-end solution with Docker/Nginx, JS, enabling real-time user audio classification to identify murmurs in heart sounds using a smartphone mic.

- **Brain Controlled VR Game** Developed a brain-computer interface using EEG (Motor Imagery + SSVEP) and EMG signals to trigger real-time gaming actions, applying CSP filtering, SVM and LSTM models for signal decoding, achieving 64% accuracy. Placed 3rd at NeuroVR Hackathon.

- **MRI Image Reconstruction** Resolved phase loss issue in TorchIO, enabling complete image reconstruction from k-space, and used it to augment motion artifacts to a motion-free MRI dataset. Trained and evaluated an MRI reconstruction algorithm (AFT-Net) on this data.

- **Drug Recommendation Application** Built a full-stack healthcare app using Flask, React, and REST APIs with real-time, personalized OTC drug recommendations, robust features (medication tracker, geolocation search, caching, OAuth), and an AI pipeline leveraging Gemini + RAG over 10,000+ MedlinePlus records, supported by scalable backend infrastructure with Redis queues and Supabase integration.

PUBLICATIONS & ACHIEVEMENTS

- Madsen, J., **Kuppa, N.**, & Parra, L. (2026). *The Brain, Body, and Behavior Dataset (BBBD): Multimodal Recordings during Educational Videos.* Published at Nature Scientific Data – **Paper Link**
- **Kuppa, N.**, Madsen, J., & Parra, L. (TBD). *Education-Attention Trap: Do viral videos ever teach what students need – A NeuroSci Approach*.
- **Kuppa, N.**, Sudhir, K., Parra, L., & Madsen, J. (TBD). *The Speed-Comprehension Tradeoff in Online Video Communication – YouTube Stats.*
- **3rd Place** – Brain-Controlled VR Game, NeuroVR Hackathon, Columbia University - **Project Link.**
- **Semifinalist -** Columbia Venture Competition with a pitch for 'Sinusitis Relief Mask'.